Tamara Ezveđ

Bioinformatics Student | Python & Java Developer

Koper, Slovenia • ezvedtamara@gmail.com • github.com/etamara6 • etamara6.github.io

EDUCATION

University of Primorska — Koper, Slovenia *Sept 2025 – Present (Expected 2028)*
Bachelor of Science in Bioinformatics

- **GPA:** 8.6 / 10.0
- **Key Coursework:** Algorithm Design, Data Structures, Programming (Java, Python), Computer Practicum (C, Linux/Bash), Mathematical Analysis, Theoretical Computer Science.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, SciPy, BioPython), Java, C, Bash/Shell Scripting.
- **Bioinformatics & CS**: De Bruijn Graphs, Eulerian Path Traversal, Sequence Alignment (NW/SW),
- **Tools & Web**: React (JavaScript/TypedArrays), Flask (REST APIs), Git, Linux CLI, LaTeX, CI/CD

SELECTED PROJECTS

De Bruijn Genome Assembler | Python, Flask, React

- Core Logic: Developed a genome assembler from scratch that reconstructs DNA sequences from short k-mer reads using De Bruijn graphs and Eulerian path traversal (Hierholzer's O(E) algorithm).
- Robust Testing: Engineered a 53-test pytest suite covering graph construction, path-finding, and assembly edge cases, ensuring 100% coverage on core logic.
- Full-Stack Tool: Built a React frontend to visualize assembly results and a Flask REST API to handle high-compute graph operations.

Sequence Alignment Studio | Python (NumPy), React, GitHub Actions

- Algorithm Engineering: Implemented Needleman-Wunsch (Global) and Smith-Waterman (Local) alignment algorithms from scratch in both Python (backend) and JavaScript (frontend fallback).
- High Performance: Leveraged NumPy vectorization and JavaScript TypedArrays to achieve sub-millisecond alignment times for typical genomic sequences.
- DevOps: Integrated GitHub Actions for automated CI, running Jest and pytest suites on every push to maintain 100% API contract reliability.

Protein Structure Visualizer | Python, BioPython, NumPy

- Scientific Tooling: Engineered a pipeline to parse PDB files and extract 3D atomic coordinates from the RCSB database for 20+ standard amino acid types.
- Computational Efficiency: Optimized data processing using NumPy vectorization and intelligent chunking (10k grid points), reducing manual property extraction time by >95%.
- Biophysical Analysis: Automated calculation of residue-level properties (GRAVY score, Instability Index, B-factor) to identify binding pockets and structural stability.

Lightkeepers Logic Engine | Java, OOP

- Designed and implemented a grid-based logic puzzle using Object-Oriented Design Patterns.
- Optimized coordinate-based propagation algorithms to ensure fluid real-time performance during complex state transitions.

LANGUAGES

- Native: Serbian, Hungarian | Fluent: English, Slovenian | Proficient: German (B1)